Chapter 22 Review

Campbell Biology, AP Edition, 12e

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Summary of Key Concepts

Does evolution proceed in a one-way direction or is it reversible? Justify your claim. (**BIG IDEA 1)**

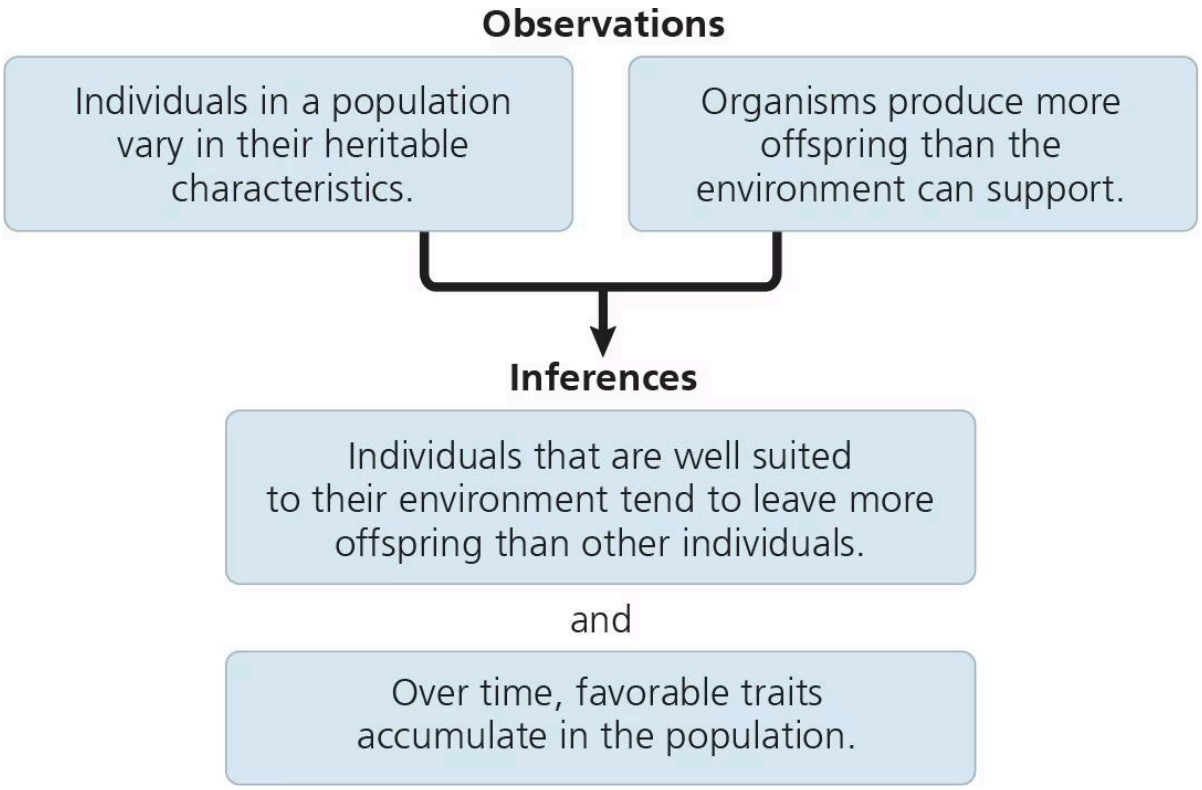
Concept 22.1: The Darwinian revolution challenged traditional views of a young Earth inhabited by unchanging species

- Darwin proposed that life's diversity arose over long periods of time from ancestral species through natural selection, a departure from prevailing views. For example, Cuvier studied ****__fossils__ but denied that ****__evolution__ occurs; he proposed that sudden catastrophic events in the past caused species to disappear from an area. Lamarck hypothesized that species evolve, but the underlying mechanisms he proposed are not supported by evidence.

Why was the age of Earth important for Darwin's ideas about evolution?

Concept 22.2: Descent with modification by natural selection explains the adaptations of organisms and the unity and diversity of life

- Darwin's experiences during the voyage of the *Beagle* gave rise to his idea that new species originate from ancestral forms through the accumulation of ****__adaptations__. He refined his theory for many years and finally published it in 1859 after learning that Wallace had come to the same idea.
- In *The Origin of Species*, Darwin proposed that over long periods of time, descent with modification produced the rich diversity of life through the mechanism of ****__natural selection__.



Flowchart showing observations. 1, individuals in a population vary in their heritable characteristics and 2, organisms produce more offspring than the environment can support. Inferences are, 1, individuals that are well suited to their environment tend to leave more offspring than other individuals, and 2, Over time, favorable traits accumulate in the population.

Describe how overreproduction and heritable variation relate to evolution by natural selection.

Concept 22.3: Evolution is supported by an overwhelming amount of scientific evidence

- Researchers have directly observed natural selection leading to evolution in many studies in diverse organisms.
- Organisms share characteristics because of common descent (****__homology__) or because natural selection affects independently evolving species in similar environments in similar ways (****__convergent evolution__).
- Fossils show that past organisms differed from living organisms, that many species have become extinct, and that species have evolved over long periods of time; fossils also document the evolutionary origin of new groups of organisms.
- Evolutionary theory can explain some ****__biogeographic__ patterns.

Summarize the different lines of evidence supporting the hypothesis that cetaceans descended from land mammals and are closely related to even-toed ungulates.